Organic Cumulative Examination

January 23, 2015

General questions on polymerization

1. Explain the concept of radical polymerization using styrene as monomer (15 points)
2. Suggest experiments to determine the molecular weight of a given polystyrene sample (15 points)
3. Suggest experiments to determine the morphology of polystyrene beads (10 points)
4. Using styrene as monomer for radical polymerizations: explain the formation of syndiotactic and isotactic polymers. What is the difference? (10 points)

Templated polymers

5. Explain the principle used to prepare templated polymers (15 points)?
6. Which kind of binding interactions are used to hold the template in place during polymerization? - give examples (15 points)
7. Imagine you intend to use a carbohydrate or amino acid as template to prepare a templated polymer. Which binding interaction would you use best and why? (10 points)
8. List at least 3 areas of current interest, where templated polymers are used or could be used (10 points)